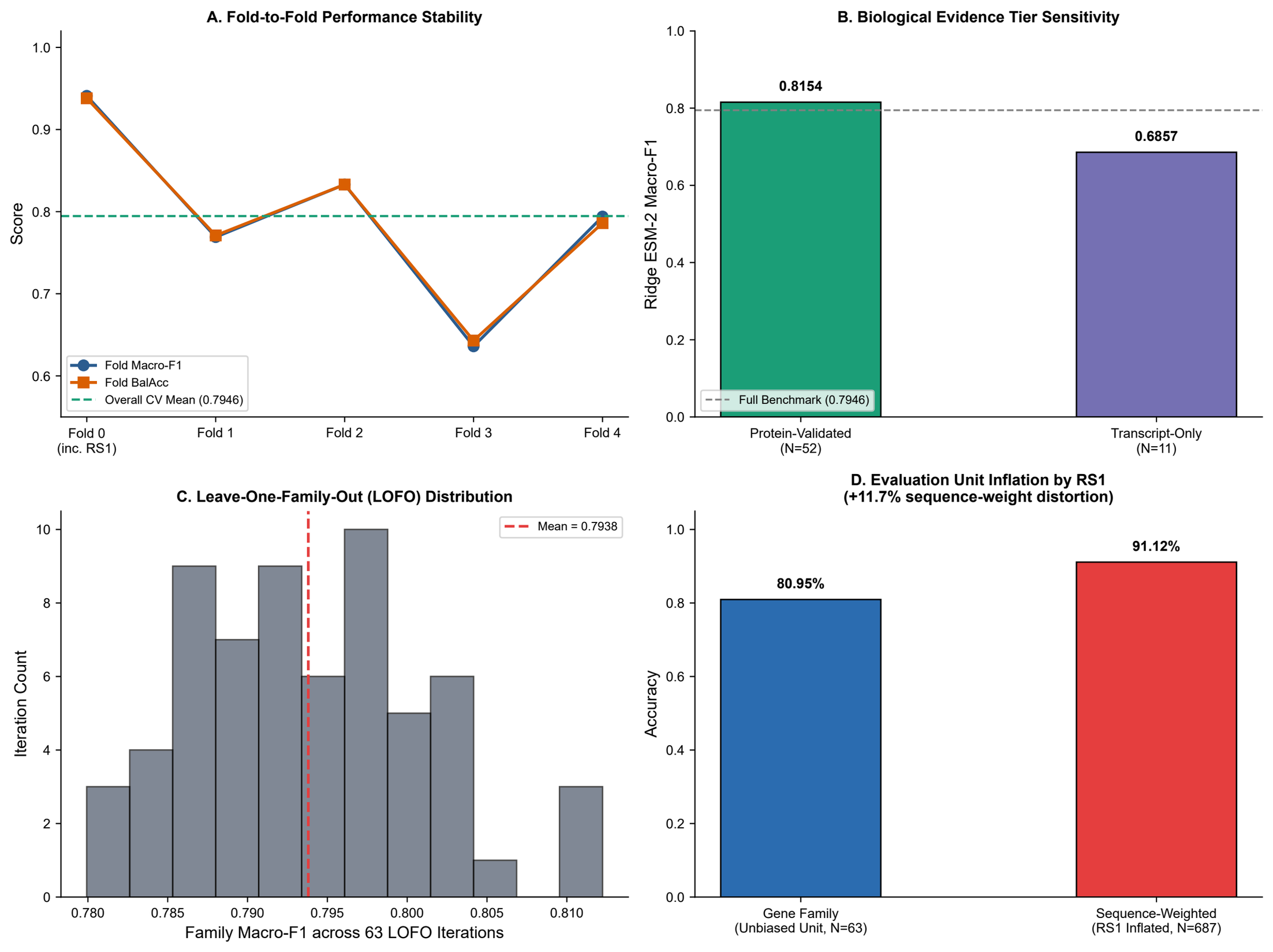

A. Fold-to-Fold Performance Stability
Score
1.0
0.9
0.8
0.7
0.6
Fold Macro-F1
Fold BalAcc
Overall CV Mean (0.7946)
Fold 0
(inc. RS1)
Fold 1
Fold 2
Fold 3
Fold 4
B. Biological Evidence Tier Sensitivity
Ridge ESM-2 Macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
0.8154
0.6857
Full Benchmark (0.7946)
Protein-Validated
(N=52)
Transcript-Only
(N=11)
C. Leave-One-Family-Out (LOFO) Distribution
Iteration Count
10
8
6
4
2
0
Mean = 0.7938
0.780
0.785
0.790
0.795
0.800
0.805
0.810
Family Macro-F1 across 63 LOFO Iterations
D. Evaluation Unit Inflation by RS1
(+11.7% sequence-weight distortion)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
80.95%
91.12%
Gene Family
(Unbiased Unit, N=63)
Sequence-Weighted
(RS1 Inflated, N=687)